INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

RECEIVER-TRANSMITTER, RADIO RT-246()/VRC AND RECEIVER-TRANSMITTER, RADIO RT-524()/VRC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of vehicular receiver-transmitters which provide short-range two-way frequency-modulation radiotelephone communication on any one of 920 channels within the frequency range of 30.00 to 75.95 megahertz (MHz). The receiver-transmitters are designated Receiver-Transmitter, Radio RT-246()/VRC and Receiver-Transmitter, Radio RT-524()/VRC. (See 6.1 and 6.7.)

1.2 Appendix. Appendix, "Production Burn-in Plan for Receiver-Transmitter, Radio RT-524()/VRC and Receiver-Transmitter, Radio RT-246()/VRC", forms a part of this specification.

2. APPLICABLE DOCUMENTS

2.1. Government documents.

2.1.1. Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703-5000, by using the Standardization Document Improvement Proposal; (DD Form 1426) appearing at the end of this document or by letter.

ASMC N/A FSC 5820

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SPECIFICATIONS

MILITARY

MIL-S-901	- Shockproof Equipment, Class HI (High Impact) Shipload
MIL-P-11268	- Parts, Materials and Processes Used in Electronic Equipment
MIL-L-12606	- Loudspeaker, Dynamic (4 inch diameter)
MIL-M-13231	- Marking of Electronic Items
MIL-I-17563	- Impregnants for Aluminum, Copper, Iron, Magnesium and Zinc Alloy Castings

STANDARDS

MILITARY

MIL-STD-252	- Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices
MIL-STD-276	- Impregnation of Porous Nonferrous Metal Castings
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	- Environmental Test Methods

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from: Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ELECTRONICS COMMAND

SC-GL-57987	- Drawing and Gage List for Receiver-Transmitter, Radio, RT-246()/VRC
SC-GL-57988	- Drawing and Gage List for Receiver-Transmitter, Radio RT-524()/VRC
SC-GL-323124	- Electrical Test Fixtures for Radio Set AN/VRC-12

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(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1. First article. When specified (see 6.2), the contractor shall furnish five (5) Receiver-Transmitter, Radio RT-524()/VRC and five (5) Receiver-transmitter, Radio RT-246()/VRC for first article inspection in accordance with the provisions of paragraph 4.3 if required in the invitation for bids or contract (see 6.4).

3.2. Construction. The equipment shall be constructed in accordance with the requirements of this specification and the applicable drawings and data lists indicated below:

Equipment	Drawing and Data List
RT-246()/VRC	SC-DL-413500
RT-524()/VRC	SC-DL-414825

3.3. Parts, materials, and processes. In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirements therein, shall apply.

3.4 Castings. Castings shall conform to requirement 21 of MIL-STD-454.

3.4.1 Quality of die castings. Quality of die castings shall be as follows (see 4.4.1).

a. Surface conditions shall be clean, dry and free from chips.

b. Cast surface finish shall be of a painting grade which allows some streaks and chill marks which can be covered and hidden by paint.

c. Flash must be removed by grinding or filing.

d. The casting after impregnation, shall be air tight, and no leakage shall be allowed when tested in accordance with 4.7.4 and 4.11.

e. Flatness shall be as specified in government-furnished drawings.

f. Dimensions shall be as specified in the government-furnished drawings.

3.4.2 Impregnation of aluminum castings. All exterior aluminum castings shall be impregnated in accordance with Standard MIL-STD-276, using Type 11 impregnating material in accordance with MIL-I-17563. Castings which have been impregnated shall be marked "IMP". The Impregnation shall be to correct

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general seepage only and shall not be used to correct poor foundry techniques or significant porosity (see 4.4).

3.5 Marking.

3.5.1 General. Marking shall conform to Specification MIL-M-13231 (see 4.4). Front panel marking shall be Group I as described in that specification.

3.5.2. Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible within minimum disassembly of the equipment.

3.5.3 Serial numbers. Receiver-Transmitter, Radio RT-524()/VRC and Receiver-Transmitter, Radio RT-246()/VRC shall be serial numbered.

3.6 Air-Seal. When the equipment is tested as specified in 4.11, the decrease in vacuum shall not exceed 0.01 pounds per square inch per minute.

3.7 Interchangeability and Gaging. Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454. Replaceable parts shall meet the requirements of 4.13. Mechanical gaging of inaccessible assemblies shall be performed prior to assembly.

3.8 Service conditions. The equipment shall meet the following service conditions:

3.8.1 Temperature.

3.8.1.1 Low. During testing as specified in 4.7.1.1, the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
	Receiver	
3.9.1.1	Muting	4.9.1.1
3.9.1.2	Loudspeaker Output	4.9.1.2
3.9.1.3	Headphone Output	4.9.1.3
3.9.1.4	Monitor Output	4.9.1.4
3.9.5	Sensitivity	4.9.5
3.9.6.1	Tone Squelch Operation	4.9.6.1
3.9.6.2	Noise Squelch Operation	4.9.6.2
3.9.7	Catching Range	4.9.7
3.9.2.1	Narrow Band Distortion	4.9.2.1
	Transmitter	
3.10.1	High Power Output	4.10.1
3.10.2	Low Power Output	4.10.2
3.10.4.1	Distortion, Narrow Band	4.10.4.1
3.10.13	Maximum Signal plus Noise-to-Noise Ratio	4.10.13
3.10.8	Automatic Frequency Selection	4.10.8

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3.10.18.1	Deviation, Narrow Band	4.10.18.1

After testing, the equipment shall meet full specification performance for these measurements.

3.8.1.2 High. During testing as specified in 4.7.1.2, the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
	Receiver	
3.9.1.1	Muting	4.9.1.1
3.9.1.2	Loudspeaker Output	4.9.1.2
3.9.1.3	Headphone Output	4.9.1.3
3.9.1.4	Monitor Output	4.9.1.4
3.9.5	Sensitivity	4.9.5
3.9.6.1	Tone Squelch Operation	4.9.6.1
3.9.6.2	Noise Squelch Operation	4.9.6.2
3.9.7	Catching Range	4.9.7
3.9.2.1	Narrow Band Distortion	4.9.2.1
	Transmitter	
3.10.1	High Power Output	4.10.1
3.10.2	Low Power Output	4.10.2
3.10.4.1	Distortion, Narrow Band	4.10.4.1
3.10.13	Maximum Signal plus Noise-to-Noise Ratio	4.10.13
3.10.8	Automatic Frequency Selection	4.10.8
3.10.18.1	Deviation, Narrow Band	4.10.18.1

After testing, the equipment shall meet full specification performance for these measurements.

3.8.2 Humidity (Moisture resistance). During testing as specified in 4.7.2, the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
	Receiver	
3.9.1.1	Muting	4.9.1.1
3.9.1.2	Loudspeaker Output	4.9.1.2
3.9.1.3	Headphone Output	4.9.1.3
3.9.1.4	Monitor Output	4.9.1.4
3.9.5	Sensitivity	4.9.5
3.9.6.1	Tone Squelch Operation	4.9.6.1
3.9.6.2	Noise Squelch Operation	4.9.6.2
3.9.7	Catching Range	4.9.7
3.9.2.1	Narrow Band Distortion	4.9.2.1
	Transmitter	
3.10.1	High Power Output	4.10.1

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3.10.2	Low Power Output	4.10.2
3.10.4.1	Distortion, Narrow Band	4.10.4.1
3.10.13	Maximum Signal plus Noise-to-Noise Ratio	4.10.13
3.10.8	Automatic Frequency Selection	4.10.8
3.10.18.1	Deviation, Narrow Band	4.10.18.1

After testing, the equipment shall meet full specification performance for these measurements.

3.8.3 Altitude (elevation). The equipment shall meet the requirements of 3.14 when tested as specified in 4.7.3.

3.8.4 Leakage (immersion). The equipment shall show no evidence of leakage after it is tested as specified in 4.7.4.

3.8.5 Orientation Test. The equipment shall meet the requirements of 3.14 when tested as specified in 4.7.5.

3.8.6 Sand and Dust. The equipment shall meet the requirements of 3.14 when tested as specified in 4.7.6.

3.8.7 Vibration.

3.8.7.1 Vibration capability. When vibrated and operated in accordance with 4.7.7.1, the receiver-transmitters shall be capable of:

a. Receiving a voice modulated signal as displayed on a Tektronic type 504 Oscilloscope or equal without break-up, loss of signal or clicks.

b. Generating and transmitting a 1000 cycle modulated rf carrier having a minimum power of 30 watts without intermittences, loss of power or oscillations as indicated on a Bird Electronic Corp Model 611 wattmeter or equal.

c. After vibration as specified in paragraph 4.7.7.2, the item shall be inspected for any structural failure, mechanical binding, or loose parts such as screws, clamps, bolts and nuts.

3.8.8 Bounce and shock. The equipment shall be capable of meeting the requirements of Table I.

3.9 Receiver electrical requirements.

3.9.1 Audio power output.

3.9.1.1 Muting. The muting level of the muting circuit for the associated loudspeaker shall be 50 Milliwatts (mW) ± 20 percent (see 4.9.1.1).

3.9.1.2 Loudspeaker output. The receiver audio amplifier shall deliver a minimum of 500 mW into a 600 ohm non-inductive load. At 22 VDC input voltage and exposure to 150 oF ambient for a minimum of one hour, loudspeaker output shall not be below a minimum of 350 mW (see 4.9.1.2).

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TABLE I. Bounce and shock.

Requirement	Test paragraph	Performance after test
Bounce	4.7.8	Specified performance (note A) no physical damage, except for paint abrasion.
Shock; ballistic	4.7.9	Specified performance (Note A) no physical damage, except for paint abrasion.

Note A: After completion of the tests, the equipment shall meet specified performance for the following: Audio (3.9.1 - less wide-band), Sensitivity (3.9.5), Squelch (3.9.6), Catching range (3.9.7), RF Power Output (3.10.1 and 3.10.2), Distortion, narrow band (3.10.4.1), Automatic frequency selection (3.10.8), Reset-ability (3.10.10), Frequency stability (3.10.11), Maximum signal plus noise to noise ratio (3.10.13), Input voltage (3.10.14), deviation (3.10.18.1)

3.9.1.3 Headphone output. The headphone output shall supply a minimum of 100 mW into a 600 ohm non-inductive load. At 22 VDC input voltage and exposure to 150° F ambient for a minimum of one hour, headphone output shall not be below a minimum of 85 mW (see 4.9.1.3).

3.9.1.4 Monitor output. The monitor output, into a 150 ohm non-inductive load, shall be 0.16 volt minimum and 0.31 volt maximum. (see 4.9.1.4).

3.9.1.5 Wide band output. The wide band circuit shall deliver 1 mW ± 1.0 db when receiving a signal having 8 kHz deviation, and 10 dbm ± 1.0 db when receiving a signal having 25 kHz deviation (see 4.9.1.5).

3.9.2 Distortion.

3.9.2.1 Narrow band distortion. Overall total narrow band distortion shall not exceed 8 percent (see 4.9.2.1).

3.9.2.2 Wide band distortion. Overall total harmonic distortion shall not exceed 5 percent with the narrow band IF filter removed and a circuit jumper substituted (see 4.9.2.2)

3.9.3 Volume control. The volume control shall simultaneously vary the loudspeaker and headphones output levels. At the minimum setting of the volume control, there shall be not more than 0.25 mW output at the loudspeaker terminal, and not more than 0.06 mW output at the headset terminal. At the maximum setting of the control, the loudspeaker and headset levels shall be not less than the levels specified in 3.9.1.2 and 3.9.1.3. Variation of the volume control settings shall maintain the monitor output level within the

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limits specified in 3.9.1.4, and the maximum variation in monitor output level shall be less than 2 db. The wide band output shall not be affected by the adjustment of the volume control (see 4.9.3).

3.9.4 Limiting. The limiting characteristic of the receiver when measured in accordance with 4.9.4 shall be such that the audio power output shall not vary more than ± 1 db (see 4.9.4).

3.9.5 Sensitivity. Rf signal inputs of 0.5 microvolts at 25.5 volts and 30.0 volts equipment inputs at room ambient temperature shall produce a signal-plus-noise-plus distortion to noise-plus-distortion ratio of 10 db minimum. Rf signal inputs of 0.5 microvolts at 22.0 volts equipment input at room ambient temperature, shall produce a S+N+D to N+D ratio of 8 db minimum. Rf signal inputs of 0.6 microvolts at -40° F and +150° F shall produce a S+N+D to N+D ratio of 10 db minimum for 25.5 volts and 30.0 volts equipment inputs and a S+N+D to N+D ratio of 8 db minimum for 22.0 volts equipment input. This requirement does not apply for frequencies of 33.90, 45.20, 56.50 and 67.80 MHz (see 4.9.5).

3.9.6 Squelch sensitivity.

3.9.6.1 Tone operation. The tone operated squelch shall open with an rf signal of 0.5 microvolt or less. At the temperature extremes (-40° F and 150° F) the tone squelch shall operate at 0.8 microvolt or less. Faulty operation of the squelch shall not occur when the incoming signal is voice modulated. The squelch once having opened, shall remain open when the rf signal is reduced to 0.4 microvolt. At the temperature extremes (-40° F and 150° F), the squelch once having opened, shall remain open when the rf signal is reduced to 0.7 microvolt. With no rf signal applied to the receiver, the maximum time for the call light to go out shall be one second, the loudspeaker output shall be zero, and the retransmit contact on the squelch relay shall be open. When the rf signal is applied, the call lamp shall light, the loud-speaker output shall appear and the retransmit contact on the squelch relay shall produce a ground (see 4.9.6.1).

3.9.6.2 Noise operation. The noise operated squelch shall open with an rf signal of 0.7 microvolt or less. At the temperature extremes (-40° F and 150° F) the noise squelch shall operate at 0.8 microvolt or less. Faulty operation of the squelch shall not occur when the incoming signal is voice modulated. The squelch, once having opened, shall remain open when the rf signal is reduced to 0.55 microvolt. At the temperature extremes, (-40° F and 150° F), the squelch, once having opened, shall remain open when the rf signal is reduced to 0.7 microvolt. With no rf signal applied to the receiver, the maximum time for the call light to go out shall be 4 seconds, the loudspeaker output shall be zero, and the retransmit contact on the squelch relay shall be open. When the rf signal is applied, the call lamp shall light, the loudspeaker output shall appear, and the retransmit contact on the squelch relay shall produce a ground (see 4.9.6.2).

3.9.7 Catching range. The sum of the plus and minus catching ranges shall not be less than 850 kHz, nor shall either the plus or minus ranges be less than 250 kHz (see 4.9.7).

3.9.8 Maximum signal plus noise to noise ratio. The maximum signal plus noise to noise ratio shall not be less than 45 db (see 4.9.8).

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3.9.9 Desensitization. A 25 millivolts interfering signal which is ± 500 kHz removed from the desired signal shall not degrade a 26 db signal plus noise to noise signal by more than 6 db (see 4.9.9).

3.9.10 Residual phase detector output. The residual phase detector output voltage shall be 0.17 vdc or less, (see 4.9.10).

3.9.11 Input voltage range. Unless otherwise specified herein, the receiver shall meet full specification requirements when operating from a 22.0 to 30.0 vdc source (see 4.9.11).

3.9.12 Audio response.

3.9.12.1 Narrow band response. The overall audio response of the receiver shall be within ± 2 db of 1000 hertz reference from 500 to 3000 hertz (see 4.9.12.1).

3.9.12.2 Wide band response. The overall audio response of the receiver shall be within + 2 db, -3 db of the 1000 Hz reference from 300 to 20,000 Hz (see 4.9.12.2)

3.9.13 VFO modulation sensitivity. The deviation sensitivity of the oscillator shall be within the limits of 300 kHz/V minimum and 600 kHz/V maximum, when measured in accordance with 4.9.13 (see 4.9.13).

3.9.14 VFO deviation capability. The deviation of the oscillator shall not be less than 450 kHz nor greater than 950 kHz over the tuning range of the oscillator, when measured in accordance with 4.9.14 (see 4.9.14).

3.9.15 VFO stability vs. temperature. The local oscillator shall not vary more than ± 180 kHz including the 85 kHz allowed for resetting. The variation shall be within ± 130 kHz if frequencies are not reset (see 4.9.15).

3.9.16 VFO resetability. When measured in accordance with 4.9.16, the dial resetability shall be such that the maximum frequency difference between any two of 5 settings to any point on the dial from either direction shall not be greater than 85 kHz. When the operation of the megahertz shorting switch is tested in accordance with 4.9.16, the minimum difference of the oscillator frequencies shall be greater than 150 kHz (see 4.9.16).

3.9.17 Output circuit impedance.

3.9.17.1 Loudspeaker output. The impedance of this circuit shall be 160 ohms ± 20 percent (see 4.9.17.1).

3.9.17.2 Headphone output. The impedance of this circuit shall be 43 ohms ± 20 percent (see 4.9.17.2).

3.9.17.3 Monitor output. The minimum output impedance of this circuit shall be 1800 ohms (see 4.9.17.3).

3.9.17.4 Wide band audio output. The output impedance of this circuit shall be 600 ohms ± 20 percent (see 4.9.17.4).

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3.9.18 IF selectivity. The IF selectivity shall conform to the following (see 4.9.18):

3.9.18.1 Narrow band.

a. Bandwidth 6 db points, 32 kHz minimum.
b. Bandwidth 60 db points, 80 kHz maximum.
c. Bandwidth 85 db points, 100 kHz maximum.

3.9.18.2 Wide band. With the narrow band filter removed and a circuit jumper substituted, IF selectivity shall be as follows:

a. Bandwidth 6 db points, 650 kHz minimum.
b. Bandwidth 40 db points, 6000 kHz maximum.

The response characteristics in the pass band shall have no irregularities greater than 2.5 db.

3.9.19 Spurious responses. There shall be no more than 10 spurious responses beyond ± 500 kHz of the desired frequency which are less than 100 db down from a one microvolt reference, and no more than 10 spurious responses within ± 500 kHz of the desired frequency which are less than 80 db down from a one microvolt reference (see 4.9.19).

3.9.20 IF rejection. If rejection shall not be less than 100 db (see 4.9.20).

3.9.21 Intermodulation. Any intermodulation resulting from two undesired carrier frequencies of the same level shall be at least 58 db down from the desired response. At the operating temperature extremes, (-40° F and 150° F), any intermodulation shall be at least 56 db down from the desired response (see 4.9.21).

3.9.22 Oscillator radiation. The radiation of any oscillator within the equipment shall produce an output not greater than 75 microvolts when the receiver is tuned to any frequency within its tuning range. This requirement does not apply for frequencies of 33.90, 45.20, 56.50 and 67.80 MHz (see 4.9.22).

3.10 Transmitter electrical requirements.

3.10.1 High power output. The rf power delivered to a 50 ohm resistive load shall be 30 watts minimum with 25.5 and 30.0 vdc input. At 22.0 vdc input, the highpower output shall be 25 watts minimum. At 25.5 vdc input, the power output shall be 50 watts maximum (see 4.10.1).

3.10.2 Low power output. The rf power delivered to a 50 ohm resistive load shall be 1/2 to 15 watts with 25.5 vdc input voltage. The low power output with 22.0 vdc input shall be 1/2 watt minimum. The low power output with 30.0 vdc input shall be 15 watts maximum (see 4.10.2).

3.10.3 Tone modulation. The 150 ± 2 Hz tone oscillator shall frequency modulate the transmitter with a resulting 3.0 kHz ± 0.5 kHz deviation. The tone shall be transmitted when the squelch switch is in the NEW OFF, NEW ON, and OLD OFF positions (see 4.10.3).

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3.10.4 Distortion.

3.10.4.1 Distortion, Narrow band. The overall total harmonic distortion shall not exceed 10 percent (see 4.10.4.1).

3.10.4.2 Distortion, wide band. The overall total harmonic distortion shall not exceed 7 percent for all modulating frequencies up to 10,000 Hz and shall not exceed 10 percent at a modulating frequency of 20,000 Hz (see 4.10.4.2).

3.10.5 Sidetone. The sidetone level at the headphone output shall be a minimum of 63 mW when the volume control is adjusted for 100 mW output at the headphone terminals. The sidetone level at the loudspeaker terminals shall be a minimum of 25 mW when the volume control is adjusted for 500 mW output at the loudspeaker terminals (see 4.10.5).

3.10.6 Modulation capability. The transmitter shall be capable of being deviated in accordance with the following, without breaking lock (see 4.10.6).

Frequency (hertz)	Deviation (kHz minimum)
500	25
1000	25
2000	25
3000	25
10000	25
20000	25

3.10.7 Catching range. The catching range of the transmitter shall be more than 750 kHz in both plus and minus directions (see 4.10.7).

3.10.8 Automatic frequency selection. The tuning time when selecting a frequency by automatic means shall not exceed 5 seconds. After presetting the pushbuttons to any of ten frequencies, no errors in frequency selection shall occur as a result of faulty operation of the automatic tuning mechanism (see 4.10.8).

3.10.9 Modulator sensitivity. The modulator sensitivity shall be in accordance with the following (see 4.10.9).

Band	Temperature	Modulator Sensitivity (kHz/V)
A	77° F	240 to 650
A	-40° F and 150° F	230 to 660
B	77° F	275 to 650
B	-40° F and 150° F	265 to 665

3.10.10 Resetability. The total backlash to the transmitter VFO shall not exceed 150 kHz (see 4.10.10).

3.10.11 Frequency stability. The frequency stability of the transmitter shall be within ± 3.5 kHz of the dial frequency (see 4.10.11).

3.10.12 Antenna switching information. The antenna switching information from the receiver-transmitter shall cause the Antenna Matching Unit

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MX-2799()/VRC of Antenna AT-912()/VRC (or the Base-Matching Unit MX-6707()/VRC of Antenna AS-1729()/VRC) to switch at the following frequencies:

33.0 MHz ± 400 kHz
37.0 MHz ± 400 kHz
42.0 MHz ± 400 kHz
47.5 MHz ± 400 kHz
53.0 MHz ± 400 kHz
56.0 MHz ± 400 kHz
60.0 MHz ± 400 kHz
65.0 MHz ± 400 kHz
70.5 MHz ± 400 kHz

If the Antenna switch has not functioned at 52.95 MHz (when testing in ascending frequency), or at 53.00 MHz (when testing in descending frequency), the unit is considered to have complied with this requirement (see 4.10.12).

3.10.13 Maximum signal plus noise to noise ratio. When measured in accordance with 4.10.13, the maximum overall signal plus noise to noise ratio of the two receiver-transmitters shall not be less than 35 db (see 4.10.13).

3.10.14 Input voltage range. The transmitter shall meet full specification requirements when operating from a 22.0 to 30.0 vdc (see 4.10.14).

3.10.15 Audio response.

3.10.15.1 Audio response, narrow band. The frequency response between 500 and 3000 hertz shall be within ± 2 db of a 1000 Hz reference, and shall be down a minimum of 14 db at 5000 Hz and above (see 4.10.15.1).

3.10.15.2 Audio response, wide band. The frequency response between 500 and 20000 Hz shall not vary by more than 3 db (see 4.10.15.2).

3.10.16 Input circuit impedance.

3.10.16.1 Narrow band impedance. The nominal impedance of this circuit shall be 150 ohms ± 10 percent at frequencies in the range of 500 to 3000 Hz (see 4.10.16.1).

3.10.16.2 Wide band impedance. The nominal impedance of this circuit shall be 600 ohms ± 10 percent at frequencies in the range of 500 to 20,000 Hz (see 4.10.16.2).

3.10.17 Spurious radiations. When measured in accordance with 4.10.17, the number of spurious outputs which exceed 10 db of quieting shall not be greater than 10.

3.10.18 Deviation.

3.10.18.1 Narrow band deviation. The nominal narrow band deviation of the transmitter shall be 8 ± kHz (see 4.10.18.1).

3.10.18.2 Wide band deviation. The nominal wide band deviation of the transmitter shall be 8 ± 2 kHz for a 1 mW signal and increase 10 ± 1 db for an applied signal of ±10 dbm (see 4.10.18.2).

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3.10.19 Deviation limiting. Deviation limiting shall be applied to the narrow band circuit only when measured in accordance with 4.10.19. The limiting shall be such that a 14 db increase (referred to the average level in 3.10.18.1) into the narrow band circuit shall increase the deviation not more than 3 db (see 4.10.19).

3.10.20 Transmitter VFO-stability vs. temperature. The transmitter VFO frequency shall not vary from its room temperature free running frequency by more than ± 250 kHz for Band A and no more then ± 300 kHz for band B, if the dial is not reset, over the specified temperature range, and an additional 150 kHz variation is permitted if the dial is reset. The frequency variation with temperature shall be within 100 kHz for Band A and 150 kHz for Band B over the specified temperature range if the frequency is not reset. Under any condition of temperature and resetting, the frequency shall not deviate from the dial setting by more than 400 kHz for Band A and 450 kHz for Band B (see 4.10.20).

3.11 Electrical requirements, Receiver-Transmitter, Radio RT-524()/VRC. This unit shall have electrical characteristics identical to Receiver-Transmitter, Radio RT-246()/VRC with the exception of the automatic frequency selection feature. A loudspeaker meeting the requirements of Specification MIL-L-12606, shall be furnished on the front panel as an integral part of the equipment. All requirements of this specification applicable to Receiver-Transmitter, Radio RT-246()/VRC, with the exception of Automatic Frequency Selection (3.10.8) are applicable to Receiver Transmitter, Radio RT-524()/VRC.

3.12 Preconditioning.

3.12.1 Bounce preconditioning. Each equipment on order shall be bounce preconditioned prior to electrical preconditioning and shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.1.

3.12.2 Electrical preconditioning. Each equipment on order shall be electrically preconditioned in accordance with Appendix A of this specification.

3.13 500-hour operational requirement. The receiver-transmitter shall be operated continuously at a 10:1 receiver-transmit ratio for a period of 500 hours at room ambient temperature and at 30 VDC input voltage, with no replacement of parts or realinement other than adjustment of front panel controls. At each 100 hour interval and at the conclusion of the 500 hour operational test, the receiver-transmitter shall meet the required tests specified in 4.14.

3.14 Operational inspection. Each equipment on order shall be given a final inspection as specified in 4.16 after all other inspections and tests are completed immediately prior to final packing and packaging.

3.15 Electromagnetic interference (EMI). The equipment shall be tested for compliance with Notice 4, MIL-STD-461. The test shall be performed on one first article sample and shall have no bearing on the acceptance of the first article sample test report (see 4.15). The following test methods are applicable: CE04, CE06, CS01, CS02, CS06, RE02, RE02.1, and RS03.2.

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3.16 Technical literature and running spare parts. Technical literature and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order (see 6.2).

3.17 Workmanship. The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

a. In MIL-P-11268G.

Paragraph	Inspection
3.8	Adhesives
3.10.7	Wires
3.15	Potting, encapsulation & embedment
3.16	Fibrous material, organic
3.21	Plastic material and parts
3.27	Finish, protective
3.30	Tropicalization of material & parts
3.31	Wiring and cabling
3.39	Controls
3.50	Lamps, lampholders, & lights
3.57	Printed wiring assemblies
3.61	Screws, other threaded devices, and related parts
3.62	Tightening of threaded parts
3.63	Lockwashers
3.64	Riveting
3.65	Brazing, soldering, and welding
3.67	Anti-seize compound
3.72	Terminals, fixed and terminal boards

b. In this specification.

Paragraph	Inspection
3.4	Castings
3.5	Marking

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements

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of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (Does not include packaging see 4.3.)

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection of equipment before packaging (see 4.5).

4.3 First article inspection. This inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, the inspection specified for Group A, Group B, and Group C (see Tables III, IV and V respectively), and 4.15. The electromagnetic interference test of 4.15 shall be performed for information only. The environmental tests shall be performed in the order shown in Table II according to the number of samples shown in the contract or order.

Table II. Order of environmental tests.

Test	Sample number				
	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5
Non-damaging tests					
Temperature (4.7.1)	1	1			
Immersion (4.7.4)			1	1	
Altitude (4.7.3)	2	2			
Orientation (4.7.5)			2	2	
Dust (4.7.6)	3	3			
Potentially damaging tests					
Humidity (4.7.2)	4	4			
Vibration (4.7.2)			3	3	
Bounce (4.7.8)			4	4	
Shock, ballistic (4.7.9)			5	5	
Electromagnetic Interference (4.15)					1

4.4. Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before packaging:

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Item	Where Required
Parts, materials & processes used in Electronic Equipment	3.3
Castings	3.4
Marking	3.5
Workmanship	3.17

4.5 Quality conformance inspection. Quality conformance inspections shall be as specified in Tables III, IV and V.

4.5.1 Quality conformance inspection of equipment before packaging. Each unit of each lot of equipment shall be subjected to Bounce and Electrical Preconditioning (see 3.12) prior to inspection. The contract shall perform the inspection specified in para 4.4 and 4.5.2 through 4.5.4.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of bounce or electrical preconditioning shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. The test equipment for government verification inspection shall be made available by the contractor.

4.5.2 Group A inspection. Each unit of each lot equipment shall be inspected for conformance to all the inspection and test requirements of Table III, group A inspection. Group A inspection shall be performed in any order which is satisfactory to the Government, except the Air-Seal Test (4.11) shall be next to last test and Operational Inspection (4.16) shall be last.

Table III. Group A inspection.

Inspection	Required paragraph	Inspection paragraph
Visual and Mechanical	3.17	4.12
A1000 VHF Tuner & Parent Board A1100 Antenna Board A1200 1st RF Board A1300 2nd RF Board A1400 Mixer Board A1500 Local Oscillator Board A4100 1st & 2nd IF Amplifier A4200 3rd, 4th & 5th IF Limiter & Discriminator A4300 Audio & Squelch Preamplifier A3100 CRS Harmonic Generator A3200 CRS Balanced Mixer A3300 CRS 2nd Mixer A3400 CRS 1st & 2nd IF Amplifier		

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Table III. Group A inspection - Continued.

Inspection	Required paragraph	Inspection paragraph
A3500 CRS 3rd If Amplifer & Limiter		
A3600 CRS Hunt Discriminator		
A3700 CRS Phase Discriminator		
A5100 Audio Amplifier		
A5200 Squelch Amplifier		
A5300 Squelch Filter		
A2100 Power Supply Board		
A2000 Crystal Switch Assembly		
A1000 RF Tuner Assembly		
A4000 Low Frequency rf Assembly		
A3000 CRS Assembly		
A5000 Audio & Squelch Assembly		
A6000 Oscillator-Buffer Case & Parent Board		
A6300 Oscillator		
A6400 Buffer		
A8100 XMTR 11.5 Modulator		
A8200 XMTR Phase Discriminator		
A8300 XMTR 1st & 2nd IF Amplifier		
A8400 XMTR Hunt Generator		
A8500 XMTR Speech Amplifer		
A7000 Null Switch Assembly (RT-246 only)		
A9000 Power Supply Assembly		
A6000 Oscillator-Buffer Assembly		
A8000 Modulator Assembly		
A7200 Servo Amplifier Assembly (RT-246 only)		
A7100 Push Button Assembly (RT-246 only)		
A6100 Driver Assembly		
A9400 Transistor Adaptor Assembly		
A6200 Power Amplifier Assembly		
FL401 Antenna Filter Assembly		
Final Front-Panel Assembly		
Assembled Receiver-Transmitter without covers		
Assembled Receiver-Transmitter with covers		
Electrical, Receiver		
Distortion, Narrow Band	3.9.2.1	4.9.2.1
Volume Control	3.9.3	4.9.3
Limiting	3.9.4	4.9.4
Input Voltage Range	3.9.11	4.9.11
Electrical, Transmitter		
Tone Modulation	3.10.3	4.10.3
Sidetone	3.10.5	4.10.5
Modulation Capability	3.10.6	4.10.6

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Table III. Group A inspection - Continued.

Inspection	Required paragraph	Inspection paragraph
Catching Range	3.10.7	4.10.7
Resetability	3.10.10	4.10.10
Input Voltage Range	3.10.14	4.10.14
Automatic Frequency Selection (RT-246 only)	3.10.8	4.10.8
Frequency Stability	3.10.11	4.10.11
Air Seal Test	3.6	4.11
Operational Inspection	3.14	4.16

4.5.3 Group B inspection. This inspection (including sampling) shall conform to Table IV. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

Table IV. Group B inspection.

Inspection	Required paragraph	Inspection paragraph
Receiver Electrical		
Maximum signal plus noise to noise ratio	3.9.8	4.9.8
Desensitization	3.9.9	4.9.9
Residual phase detector output	3.9.10	4.9.10
Audio response, narrow band	3.9.12.1	4.9.12.1
VFO modulating sensitivity	3.9.13	4.9.13
VFO deviation capability	3.9.14	4.9.14
VFO stability vs temperature	3.9.15	4.9.15
VFO resetability	3.9.16	4.9.16
Transmitter Electrical		
Modulator sensitivity	3.10.9	4.10.9
Frequency stability	3.10.11	4.10.11
Antenna switching information	3.10.12	4.10.12
Interchangeability (dimensions)	3.7	4.13
Interchangeability (module performance)	3.7	4.13

4.5.3.1 Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

4.5.3.2 Group B sampling plan. The group B sampling is as shown in Table V.

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Table V. Group B sampling plan.

Electrical		Inspection by gages (mechanical)	
Lot size	Sample size,	Dimensions Sample size	Module Performance Sample size
2 to 8	3	2	5
9 to 15	3	2	5
16 to 25	3	3	5
26 to 50	5	5	5
51 to 90	6	5	7
91 to 150	7	6	11
151 to 280	10	7	13
281 to 500	11	9	16
501 to 1200	15	11	19
1201 to 3200	18	13	23
3201 to 10000	22	15	29

NOTE: Acceptance number in all cases is zero.

4.5.4 Group C inspection. This inspection shall consist of the tests specified in Table VI and shall be performed on sample units that have been subjected to and met Groups A and B inspection.

Table VI. Group C inspection.

Inspection	Required paragraph	Inspection paragraph
Subgroup I		
Immersion	3.8.4	4.7.4
Wideband tests, receiver:		
Audio power output, wide band output	3.9.1.5	4.9.1.5
Distortion, wide band	3.9.2.2	4.9.2.2
Audio response, wide band	3.9.12.2	4.9.12.2
Wide band output impedance	3.9.17.4	4.9.17.4
IF selectivity, wide band	3.9.18.2	4.9.18
Wideband tests, transmitter:		
Distortion, wide band	3.10.4.2	4.10.4.2
Audio response, wide band	3.10.15.2	4.10.15.2
Wide band circuit impedance	3.10.16.2	4.10.16.2
Deviation, wide band	3.10.18.2	4.10.18.2
Subgroup III		
Loudspeaker output impedance	3.9.17.1	4.9.17.1
Headphone output impedance	3.9.17.2	4.9.17.2
Monitor output impedance	3.9.17.3	4.9.17.3
Narrow band circuit impedance	3.10.16.1	4.10.16.1

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Table VI. Group C inspection - Continued.

Inspection	Required paragraph	Inspection paragraph
Subgroup IV		
Receiver:		
IF selectivity, narrow band	3.9.18.1	4.9.18
Spurious responses	3.9.19	4.9.19
IF rejection	3.9.20	4.9.20
Intermodulation	3.9.21	4.9.21
Oscillator radiation	3.9.22	4.9.22
Transmitter:		
response, narrow band	3.10.15.1	4.10.15.1
Spurious radiation	3.10.17	4.10.17
Deviation limiting	3.10.19	4.10.19
VFO stability	3.10.20	4.10.20
Subgroup V		
500 hour operational test	3.13	4.14
Bounce	3.8.8	4.7.8
Temperature	3.8.1	4.7.1
Vibration, capability	3.8.7.2	4.7.7.2
Subgroup VI		
Humidity (moisture resistance)	3.8.2	4.7.2
Subgroup VII		
Altitude	3.8.3	4.7.3
Orientation	3.8.5	4.7.5
Sand and dust	3.8.6	4.7.6
Vibration, internal	3.8.7.1	4.7.7.1
Shock, ballistic	3.8.8	4.7.9

4.5.4.1 Sampling for inspection of equipment. Units selected for each Group C inspection shall be selected without regard to their quality in accordance with the following:

4.5.4.1.1 Subgroups I, III, and IV. For these subgroups, two units from the first production lot shall be selected for the first Group C inspection. For subsequent Group C inspection, two units from each successive 100, or two per month if less than 100 are produced monthly, shall be selected.

4.5.4.1.2 Subgroup V. For this subgroup, two units from the first production lot shall be selected for the first Group C inspection. For subsequent Group C inspections, two units per month shall be selected.

4.5.4.1.3 Subgroup VI. For this subgroup, two units from the first production lot shall be selected for the first Group C inspection. For subsequent Group C inspections, two units every two months shall be selected.

4.5.4.1.4 Subgroup VII. For this subgroup, two units from the first

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production lot shall be selected for the first Group C inspection. For subsequent Group C inspections, two units every six months shall be selected.

4.5.4.2 Noncompliance. The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and, (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.5 Reinspection of conforming group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Preconditioning.

4.6.1 Bounce preconditioning. The receiver-transmitter shall then be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute (see 3.12.1).

4.6.2 Electrical preconditioning. All Receivers-Transmitters, RT-246()/VRC and RT-524()/VRC, on order shall be electrically preconditioned in accordance with the burn-in plan delineated in Appendix A of this specification (see 3.12.2).

4.7 Service conditions tests.

4.7.1 Temperature test.

4.7.1.1 Low temperature. The equipment shall be tested in accordance with Method 502, Procedure I, of MIL-STD-810. The storage temperature shall be -80 oF and the operating temperature shall be -40 oF. Prior to testing, the equipment shall fully comply with the performance requirements specified in 3.8.1.1.

4.7.1.2 High temperature. The equipment shall be tested in accordance with Method 501, Procedure III, of MIL-STD-810. The operating temperature shall be +150 oF. Prior to testing, the equipment shall fully comply with the performance requirements specified in 3.8.1.2.

4.7.2 Humidity (moisture-resistance) test. The equipment shall be tested in

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accordance with Method 507, Procedure III, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.8.2. Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.

4.7.3 Altitude test. The receiver-transmitter mounted on Mounting MT-1029()/VRC shall be tested in accordance with Method 500, Procedure 1, of MIL-STD-810. The operational test of 4.16 shall be performed before, during, and after the altitude test to determine compliance with 3.8.3.

4.7.4 Leakage (Immersion) test. The equipment, as prepared for field transportation, shall be tested in accordance with Method 512, Procedure 1, of MIL-STD-810, to determine compliance with 3.8.4.

4.7.5 Orientation test. The equipment shall be inclined for a minimum of 5 minutes in each plane (forward, backward, left, and right) to an angle of 20 (+3, -1) degrees. During inclination in each plane, the equipment shall be tested to 4.16 to determine compliance with 3.8.5.

4.7.6 Sand and dust test. The equipment shall be tested in accordance with Method 510, Procedure I, of MIL-STD-810. The operational test of 4.16 shall be performed before, during, and after dust testing to determine compliance with 3.8.6.

4.7.7 Vibration.

4.7.7.1 Test for vibration capability. Vibration of the equipment shall be measured as follows, to determine conformance to 3.8.7.1.

a. Secure the equipment directly to a vibration table that can be controlled within 10 percent of the specified amplitude.

b. Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be not less than 0.030 inch. Vibrate the equipment by sweeping at a constant rate from 10 to 55 cycles per second and then from 55 to 10 cycles per second. The total sweep up and down shall take 7 1/2 minutes per plane.

c. Performance. To determine receive and transmit capability, during the vibration test, the equipment shall be operated in receive and transmit modes (9 to 1 duty cycle) over the vibration frequency range specified in 4.7.7.1b for all three directions to determine compliance with 3.8.7.1.

4.7.8 Bounce vehicular. The receiver-transmitter mounted on Mounting MT-1029()/VRC shall be tested in accordance with Method 514, Procedure IX, Part 2 of MIL-STD-810. During the bounce test, the receiver shall be operated

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under the standard test conditions of paragraph 4.8. After completion of the bounce test, the equipment shall be tested to determine compliance with 3.8.8.

4.7.9 Shock test; ballistic. The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in specification MIL-S-901. The equipment, mounted on Mounting MT-1029()/VRC, shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of 9 blows: one each 1-foot blow, 3-feet blow, and 5-foot blow on the back, side, and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible. After completion of the shock test, the equipment shall be tested to determine compliance with 3.8.8.

4.8 Standard test conditions. All electrical tests shall be performed as follows:

a. In a screened room conforming to Specification MIL-STD-461. Testing of the transmitter section of the receiver-transmitter need not be performed in a screened room provided proper shielding techniques are used which will minimize radiation to the extent that the accuracy of any measurements required by this specification are not affected. However, if the Government finds that inadequate shielding techniques are being used, screened rooms shall be provided by the contractor for these tests.

b. The electrical tests specified in 4.7.1, 4.7.2, 4.7.3, 4.7.5, 4.7.6, 4.7.9, 4.7.10, and 4.7.11 shall be performed in an interference free area, with necessary precautions taken to prevent ambient interference from entering or affecting either the measuring circuit or equipments under test.

c. Under the following standard test conditions, unless otherwise specified:

Receiving

Temperature	Room ambient
Humidity	Room ambient
Input supply voltage	25.5 vdc
RF input	10 microvolts, 8 kHz deviation at 1 kHz modulation frequency.
Test frequency	60.05 Megahertz
Squelch	Old "OFF"
Audio output level:	
Loudspeaker	500 mW
Headphone	100 mW
Wideband	0 dbm
Audio load impedance	600 ohms
Narrow band IF filter	

Transmitting

Temperature	Room ambient
Humidity	Room ambient

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Input supply voltage	25.5 vdc
Audio input voltages:	
Narrow band circuit	2.8 millivolts rms open circuit voltage at 1000 Hz using a generator having an internal impedance of 150 ohms.
Wide band circuit	1.55 volts rms open circuit at 1000 Hz using a generator having an internal impedance of 600 ohms.
Dummy antenna load	50 ohms resistive
Tone generator (150 Hz)	Disabled (old squelch "ON")
Test frequency	40.10 Megahertz
RF power output	Low power

4.9 Electrical tests, receiver.

4.9.1 Audio output power.

4.9.1.1 Muting. The receiver shall be operated under standard test conditions with the Volume Control set for 500 mW speaker output. When the muting circuit is energized, the loudspeaker lever shall decrease to 50 mW ± 20 percent (see 3.9.1.1).

4.9.1.2 Loudspeaker output. With the volume control set fully clockwise, the power dissipated into a 600 ohm non-inductive load shall be measured to determine compliance with 3.9.1.2.

4.9.1.3 Headphone output. With the volume control set fully clockwise, the power dissipated into a 600 ohm non-inductive load shall be measured to determine compliance with 3.9.1.3.

4.9.1.4 Monitor amplifier output. With the volume control set for 500 mW speaker output, and the monitor amplifier terminated into a 150 ohm non-inductive lead, the output voltage shall be measured to determine compliance with 3.9.1.4

4.9.1.5 Wide band output. With the rf input at 200 uv, the receiver shall be operated under standard test conditions except the narrow band IF filter shall be removed and replaced with a dummy filter having the input and output tied together internally. The deviation shall be as specified herein. The power output shall be measured for deviations of 8 and 25 kHz to determine compliance with 3.9.1.5.

4.9.2 Distortion.

4.9.2.1 Distortion, narrow band. The receiver shall be operated under standard test conditions. The distortion at the loudspeaker terminal shall be measured to determine compliance with 3.9.2.1.

4.9.2.2 Distortion, wide band. The receiver shall be operated under standard

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test conditions except the narrow band IF filter shall be removed and replaced with a dummy filter having the input and output connected together internally. The rf input to the receiver shall be 200 uv. The distortion at the wide band audio output shall be measured at 1000, 5000, and 10,000 Hertz to determine compliance with 3.9.2.2.

4.9.3 Volume control. The receiver shall be operated under standard test conditions. The output of the loudspeaker, headphone, monitor, and wide band circuits shall be measured for full clockwise (cw) and counter clockwise (ccw) settings of the volume control to determine compliance with 3.9.3.

4.9.4 Limiting. The receiver shall be operated under standard test conditions except that the rf input shall be 1.0 microvolt. The signal generator output shall then be increased to 100,000 microvolts and the audio power output continuously observed. The test shall be conducted at frequencies of 30.00, 75.95, 65.55, 41.05, 53.00 and 52.10 megahertz, and in that order. The receiver shall comply with 3.9.4.

4.9.5 Sensitivity. The receiver shall be operated under standard test conditions except the rf input shall be 0.5 microvolt. The ratio of the signal plus noise plus distortion to noise plus distortion, expressed in db, shall be measured with an appropriate distortion analyzer. Measurements shall be made at frequencies of 30.00, 65.10, 41.05, 75.90, 52.95, 53.00, 41.50, and 64.50 megahertz, in that order. At the temperature extremes (-40° F and 150° F), this test shall be performed with an rf input of 0.6 microvolts. Sensitivity shall comply with 3.9.5.

4.9.6 Squelch sensitivity.

4.9.6.1 Tone operation. The receiver shall be operated under standard test conditions except the rf input shall be 0.5 microvolt. The squelch shall be adjusted to the NEW SQUELCH ON position, and the signal generator modulated with an additional tone of 150 Hz, 3 kHz deviation. The rf signal shall be applied to the receiver and then removed from the receiver to determine compliance with 3.9.6.1. The test shall be conducted at frequencies of 30.00, 65.20, 41.30, 52.95, 53.00 and 75.95 megahertz, in that order.

4.9.6.2 Noise operation. The receiver shall be operated under standard test conditions except the rf input shall be 0.7 microvolt and the squelch control adjusted to the OLD SQUELCH ON position. The rf signal shall be applied to the receiver and then removed from the receiver to determine compliance with 3.9.6.2. The test shall be conducted at frequencies of 30.00, 65.40, 41.50, 53.60, 52.70 and 75.80 megahertz, in that order.

4.9.7 Catching range. The receiver shall be operated under standard test conditions. Disconnect all cables from the crystal switch assembly and connect to a crystal switch assembly in another receiver or to a test jig containing a crystal switch assembly. Tune the receiver under test to 30.90 MHz and the crystal switch to the appropriate corresponding frequency. Tune the dial of the receiver under test in 50 kHz increments higher in frequency. After each movement of the dial, momentarily disable (short) the control line and observe whether or not the test signal can be received. Record the last dial reading at which the receiver is capable of receiving the 30.90 MHz signal. Repeat this procedure by setting the mc dial to 31.90 MHz and moving the dial of the receiver under test in 50 kHz increments lower in frequency.

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This test shall be repeated at each of the following pairs of frequencies to determine compliance with 3.9.7.

(30.90, 31.90), (31.95, 32.95), (32.90, 33.90), (33.95, 34.95),
(34.90, 35.90), (35.95, 36.95), (36.90, 37.90), (37.95, 38.95),
(38.90, 39.90), (39.95, 40.95), (40.90, 41.90), (41.95, 42.95),
(42.90, 43.90), (43.95, 44.95), (44.90, 45.90), (45.95, 46.95),
(46.90, 47.90), (47.95, 48.95), (48.90, 49.90), (49.95, 50.95),
(50.90, 51.90), (51.95, 52.95),

4.9.8 Maximum signal plus noise to noise ratio. The receiver shall be operated under standard test conditions except the rf input shall be 100 microvolts. The ratio, expressed in db, of the audio output with and without modulation shall be measured. The test shall be conducted at frequencies of 30.00, 65.70, 52.95, 75.85, 45.80, and 53.10 megahertz, in that order, to determine compliance with 3.9.8.

4.9.9 Desensitization. Apply an rf signal to the receiver at a level to produce a 26 db S+N/N ratio. An interfering signal which is + 0.5 MHz removed from the desired rf signal shall be applied, and its level increased until the S+N/N ratio of the desired signal is reduced to 20 db. The level of the interfering signal shall be recorded to determine compliance with 3.9.9. The test shall be conducted at frequencies of 30.00, 52.95, 53.05, and 75.10 megahertz.

4.9.10 Residual phase detection output. The control loop shall be open (error signal lead connected). Connect a dc VTVM so that the output of the phase detector only is measured. With the receiver dial set to 41.45 MHz and with a 250 millivolts rf signal applied into the crystal Reference System high pass filter, measure the residual dc output voltage while varying the rf signal frequency from 52.55 to 53.35 MHz. Disregard the output reading at 52.95 MHz. The residual phase detector output shall comply with 3.9.10.

4.9.11 Input voltage tests. To determine conformance with 3.9.11, the tests listed below shall be conducted at input voltages of 22.0 and 30.0 vdc. Automatic frequency selection (RT-246()/VRC only) shall be conducted at input voltages of 25.0 and 27.0 vdc.

a. Audio output power (see 4.9.1.1, 4.9.1.2, 4.9.1.3 and 4.9.1.4).

b. Sensitivity (see 4.9.5).

c. Squelch sensitivity (see 4.9.6).

d. Catching range (see 4.9.7).

e. Distortion (see 4.9.2.1)

f. Automatic frequency selection (see 4.10.8).

4.9.12 Audio response.

4.9.12.1 Audio response, narrow band. The receiver shall be operated under standard test conditions. The signal generator shall be modulated with an external audio oscillator and the deviation set to 8 kHz at 1000 Hz modulated

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frequency. The audio output of the receiver shall be measured for frequencies of 500, 1000, 2000, and 3000 Hz, to determine compliance with 3.9.12.1.

4.9.12.2 Audio response, wide band. The receiver shall be operated at standard test conditions except the wide band IF system shall be installed. The test shall be performed in the same manner as that of 4.9.12.1, except measurements shall be made at frequencies of 300, 1000, 3000, 5000, 10,000, and 20,000 Hz, to determine compliance with 3.9.12.2.

4.9.13 VFO modulation sensitivity. The deviation of the local oscillator from the nominal frequency shall be measured when + 0.5 vdc is applied to the modulator input through a 250k ohm resistor. With the control line open (no voltage applied to the modulator input), measure the VFO free running frequency. These measurements shall be taken at nominal VFO frequencies of 42.00 MHz,(53.50 MHz dial frequency), 52.00 MHz (63.50 MHz dial frequency), and 63.00 MHz (74.50 MHz dial frequency) to determine compliance with 3.9.13.

4.9.14 VFO deviation capability. The deviation of the VFO from nominal frequency shall be measured when ± 1.5 vdc is applied to the modulator input through a 250k ohm resistor. With the control line open (no voltage applied to the modulator input), measure the VFO free running frequency. These measurements shall be taken at nominal VFO frequencies of 42.00 MHz (53.50 MHz dial frequency), 52.00 MHz (63.50 MHz dial frequency), and 63.00 MHz (74.50 MHz dial frequency) to determine compliance with 3.9.14.

4.9.15 VFO stability vs temperature. The automatic frequency control line of the VFO shall be grounded and the nominal frequency of the VFO measured at room temperature. The ambient temperature shall then be lowered to -40° C, and maintained at that temperature for one hour, and then the VFO frequency shall be measured. The ambient temperature shall then be raised to +65° C, and maintained for one hour, and the nominal VFO frequency shall be measured. At each temperature step, the nominal VFO frequency shall be measured at dial frequencies, of 53.50, 63.50, and 74.50 MHz. This test shall be conducted at input voltages of 22.0, 25.5, and 30.0 volts dc. Temperature stability shall comply with 3.9.15.

4.9.16 VFO resetability. With the VFO control line grounded and no input, turn the kHz and MHz knobs at least 5 detents below the test frequency where possible. The kHz knob shall be turned up to .50 MHz followed by turning the MHz knob up to 31 MHz to give the test frequency of 31.50 MHz. The frequency of the VFO shall be measured. Repeat the above, except the receiver shall be tuned to 31.50 MHz from the direction of a higher frequency. This procedure shall be repeated five times, from each direction. The test shall be repeated for dial frequencies of 41.50 and 51.50 MHz to determine compliance with 3.9.16. At each test frequency, the operations of the megacycle shorting switch shall be tested with a dc VTVM connected between TP 3002 and ground. The MHz knob shall be seated by rocking slightly, and the oscillator frequency recorded. The MHz knob shall be stressed in a cw direction until the shorting switch is actuated as indicated by a sudden increase to 16 vdc on the VTVM. AT this knob position, the oscillator frequency shall be measured. The difference between the oscillator frequencies shall conform to 3.9.16. Repeat in a ccw direction at each test frequency.

4.9.17 Output circuit impedance. These tests shall be conducted under standard test conditions.

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4.9.17.1 Loudspeaker output impedance. The loudspeaker output shall be terminated in 160 ohms and the audio voltage across the termination adjusted with the volume control to one (1) volt. When the load is removed, the open circuit voltage shall increase to 2 ± 0.4 volts (see 3.9.17.1).

4.9.17.2 Headphone output impedance. The headphone output shall be terminated in 43 ohms and the audio voltage across the termination adjusted with the volume control to 0.5 volt. When the load is removed, the open circuit output voltage shall increase to 1 ± 0.2 volts (see 3.9.17.2).

4.9.17.3 Monitor output impedance. For this test, the 8 kHz deviation shall be adjusted for a convenient unclipped reference level voltage at the unterminated monitor output and the reference level voltage shall be measured. Terminate the monitor output with a load of 1800 ohms. The open circuit voltage shall be at least twice the loaded value (see 3.9.17.3).

4.9.17.4 Wide band output impedance. The wide band output shall be terminated in 600 ohms and the output voltage measured. With the load removed, the open circuit voltage shall be twice the loaded value ± 10 percent (see 3.9.17.4).

4.9.18 IF selectivity, narrow band or wide band. A signal shall be injected into the input of the IF and the rectified signal at the output of the fourth stage of the IF shall be monitored. Adjust the input until this signal is at least 6 db below saturation. With this signal level as a reference, adjust the input frequency and level to determine compliance with 3.9.18.1 and 3.9.18.2.

4.9.19 Spurious response. The amount of quieting produced by an unmodulated rf carrier of 1.0 microvolt shall be used as a reference. The signal generator output shall be increased to 100,000 microvolts and the generator tuned throughout the range of 20 to 100 MHz. Those frequencies which produce the quieting equal to or greater than the reference shall be recorded to determine compliance with 3.9.19. The test shall be conducted at frequencies of 30.00 and 69.25 MHz.

4.9.20 IF rejection. The receiver shall be tuned to 30.00 MHz. The amount of quieting produced by 1 microvolt shall be used as a reference. The signal generator shall then be tuned to 11.5 MHz and its output raised until the amount of quieting is equal to the references. The number of microvolts shall be recorded to determine compliance with 3.9.20. The test shall be performed at frequencies of 30.00, 52.00, 54.00, and 75.00 MHz.

4.9.21 Intermodulation. The receiver shall be tuned to 41.5 MHz and on unmodulated rf carrier of 1.0 microvolt shall be applied to the input of the receiver. The amount of quieting produced by this signal shall be used as a reference. Two generators, similar in all respects, tuned to 41.6 and 41.7 MHz shall be coupled to the input of the receiver. The output of the generators shall be kept equal and simultaneously increased until the reference quieting is produced. The number of microvolts required to reproduce the reference shall be recorded to determine compliance with 3.9.21. The loading effect of the generators shall be considered when arriving at the number of microvolts required to produce the reference.

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4.9.22 Oscillator radiation. The receiver shall be operated under standard test conditions except that no input signal shall be applied. A field strength meter, or equivalent, shall be connected to the antenna terminals of the receiver. The receiver shall be tuned to 30.00 MHz and scanned with the field strength meter. The frequencies and amplitudes of detectable outputs greater than 50 microvolts shall be recorded to determine compliance with 3.9.22. This procedure shall be repeated for dial settings of 46.50, 53.20, and 75.60 MHz.

4.10 Electrical tests, transmitter.

4.10.1 High power output. The transmitter shall be operated at standard test conditions except that no modulation shall be applied. With the power switch set to "HI", the power output shall be measured at frequencies of 30.00, 65.20, 41.05, 75.95, 52.90, and 53.00 MHz, in that order. The equipment shall comply with 3.10.1.

4.10.2 Low power output. The test of 4.10.1 shall be repeated except the power switch shall be set to "LO", to determine compliance with 3.10.2.

4.10.3 Tone modulation. The transmitter shall be operated under standard test conditions into a deviation meter through a 50 db pad, except that the 150 hertz tone modulator shall not be disabled. The tone shall be transmitted when the squelch switch is set to "NEW OFF", "NEW ON", and "OLD OFF" positions and the deviation and frequency of the 150 hertz tone shall be measured to determine compliance with 3.10.3.

4.10.4 Distortion. These tests shall be conducted with two receiver-transmitters, one transmitting, the other receiving. An approved unit shall be used in the receiver position. The rf signal level to the receiver shall not be less than 50 microvolts.

4.10.4.1 Distortion, narrow band. With both receiver and transmitter operating under the appropriate standard test conditions, the total harmonic distortion measured at the audio output jack of the approved receiver-transmitter with 1000 Hz applied, shall meet the requirements of 3.10.4.1.

4.10.4.2 Distortion, wide band. With the transmitter deviated by a signal applied at the wide band terminal producing a 20 kHz deviation, the distortion shall be measured at the wide band output of a receiver which has the narrow band filters removed and replaced with dummy filters having input and output connected together internally. The transmitter output shall be attenuated to give an rf input to the receiver between 500 and 10,000 microvolts. Distortion measurements shall be made at 1000, 5000, 10,000 and 20,000 hertz to determine compliance with 3.10.4.2.

4.10.5 Sidetone. The volume control shall be adjusted for standard "receive" conditions. The transmitter shall then be operated under standard test conditions and standard modulation applied to the narrow band only. The output of the headphone terminals and loudspeaker terminals shall be measured to determine compliance with 3.10.5.

4.10.6 Modulation capability. With an approved receiver tuned to the transmitter frequency of the receiver-transmitter under test, the transmitter

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shall be modulated at 500, 1000, 2000, 3000, 10,000 and 20,000 hertz with 2.5 volts minimum applied to the "X-MODE" input. At each modulation frequency the input shall be increased until the transmitter breaks lock. The maximum deviation reading before the transmitter breaks lock shall meet the requirements of 3.10.6. The test shall be conducted at transmitter dial frequencies of 35.00, 48.00, 57.00, and 70.00 MHz.

4.10.7 Catching range. A test unit consisting of an approved receiver tuner and crystal unit shall be used for this test. P-1002 (rf output from transmitter buffer) and P-1003 (IF output) shall be substituted for the equivalent connections in the transmitter. The transmitter shall be set to a dial frequency of 35.00 MHz and the test unit set on the compatible frequency. The receiver shall be tuned to the transmitter frequency and the tone modulation (1000 hertz at 1.4 millivolts to the MIKE input) of the transmitter monitored. The transmitter shall then be detuned lower in frequency (minus catching range) than the original setting until the lock is broken, indicated by a loss of signal at the monitoring equipment. The transmitter shall then be tuned towards its original setting in 50 kHz increments until the signal at the monitoring receiver is regained. Record the difference between the original signal and the setting when the signal is regained. The procedure shall be repeated by detuning the transmitter higher in frequency, to determine the plus catching range. Repeat the above for dial frequencies of 48.00, 57.00, and 70.00 MHz. The catching range shall comply with 3.10.7.

4.10.8 Automatic frequency selection. With the receiver-transmitter set for automatic operation adjust any one pushbutton potentiometer to 30.00 MHz and any other pushbutton to 52.95 MHz. Record the maximum tuning time for the SERVO to cycle from 30.00 MHz up to 52.95 MHz and the time to cycle from 52.95 MHz down to 30.00 MHz. All pushbuttons shall then be set to five (5) different frequencies in the "A" band, five (5) different frequencies in the "B" band and the pushbutton potentiometers checked for resetability by approaching from both sides of the pushbutton's setting, to determine compliance with 3.10.8.

4.10.9 Modulator sensitivity. With the output of the transmitter connected to a frequency counter, the VFO control line shall be opened and + 1.0 vdc shall be applied to the VFO jack and the resultant frequencies shall be measured and recorded. The difference between the two resultant frequencies in kHz, divided by 2, is the modulator sensitivity in kHz/V. The test shall be conducted at transmitter frequencies of 30.00, 45.00, 52.00, 53.00, 68.00, and 75.00 MHz to determine compliance with 3.10.9.

4.10.10 Resetability. With the VFO control line open and the VFO jack terminated to ground through a 50 ohm resistor, the receiver-transmitter shall be operated with no rf input, into a frequency counter through a 50 ohm variable pad. The VFO frequency shall be measured after approaching the test frequency from both a clockwise and a counterclockwise direction. This shall be done by setting the kHz and MHz knobs five detents below the test frequency where possible and then setting the kHz knob to frequency followed by setting the MHz knob to frequency. After measuring VFO frequency, the test frequency shall then be approached from a higher frequency, in the same manner. The difference between the two VFO frequencies shall be considered the resetability. The test shall be conducted at test frequencies of 41.50, 51.50, 54.50, 64.50, and 74.50 MHz to determine compliance with 3.10.10.

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4.10.11 Frequency stability. The output of the transmitter shall be attenuated to the proper level and fed into a frequency counter that has been properly calibrated. The transmitter output frequency shall be measured at the following dial frequencies:

30.00 through 30.95 MHz
41.00 through 41.95 MHz
51.00 through 53.95 MHz
64.00 through 64.95 MHz
75.00 through 75.95 MHz

The tests shall first be performed in an increasing order of the frequencies shown and then in a decreasing order. The transmitted frequency shall comply with 3.10.11.

4.10.12 Antenna information (switching). With the transmitter connected to the Antenna Matching Unit MX-2799()/VRC or MX-6707()/VRC, the transmitter shall be tuned over the frequency range of both bands (A and B), and the point at which the matching unit switches bands noted and recorded. Repeat, approaching the frequencies from the opposite direction. The switching point shall conform to the requirements of 3.10.12. For production testing, a test jig or fixture simulating Antenna Matching Unit MX-2799 ()/VRC or Matching Unit Base MX-6707()/VRC shall be utilized.

4.10.13 Maximum signal plus noise to noise ratio. The transmitter shall be operated under standard test conditions with modulation of 400 millivolts at 1000 Hz applied to the Remote Mike input. The transmitter dial frequency shall be 75.95 MHz. An approved receiver-transmitter shall be tuned to the transmitter frequency and the S + N/N ratio shall be measured to determine compliance with 3.10.13. The level of the signal received by the approved receiver-transmitter shall be 100 microvolts minimum.

4.10.14 Input voltage range tests. To determine compliance with 3.10.14, the tests listed below shall be conducted at input voltages of 22.0 and 30.0 vdc:

a. Power output, high and low (4.10.1 and 4.10.2)

b. Distortion, narrow band (4.10.4.1)

c. Maximum signal plus noise to noise ratio ((4.10.13)

d. Deviation, narrow band (4.10.18.1)

4.10.15 Audio response.

4.10.15.1 Audio response narrow band. The transmitter shall be operated under standard test conditions. The transmitter shall be set to 30.00 MHz and the deviation measured for the following modulating frequencies: 500, 1000, 5000, 10,000 and 20,000 hertz. The ratio expressed in db, of the deviation at various frequencies to that at 1000 hertz, shall be recorded, and shall comply with 3.10.15.1. The test shall be repeated at 54.00 MHz.

4.10.15.2 Audio response, wide band. The transmitter shall be set to 30.00 MHz and the deviation measured for the following modulating frequencies: 500, 1000, 5000, 10,000 20,000 hertz. The ratio expressed in db, of the deviation

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at various frequencies to the deviation at 1000 Hz shall be recorded, and shall comply with 3.10.15.2. The test shall be repeated at 54.00 MHz.

4.10.16 Input circuit impedance.

4.10.16.1 Narrow band impedance. The narrow band circuit impedance shall be measured at 1000 Hz to determine compliance with 3.10.16.1.

4.10.16.2 Wide band impedance. The wide band circuit impedance shall be measured at 1000 Hz to determine compliance with 3.10 16.2.

4.10.17 Spurious radiation. The transmitter shall be operated under standard test conditions except that no modulation shall be applied. The output of the transmitter shall be connected to an approved receiver through a 50 ohm variable pad to provide 70 millivolts across the receiver antenna terminals. With the transmitter on any one of the frequencies listed below, the receiver shall be tuned throughout its tuning range and points of quieting which exceed 10 db shall be recorded to determine compliance with 3.10.17. The test shall be conducted at transmitter frequencies of 34.50, 46.00, 57.50, and 69.00 MHz. To distinguish between receiver and transmitter spurious radiation; quieting points shall be checked with a signal generator, tuned to frequency of quieting, substituted in place of the transmitter.

4.10.18 Deviation test.

4.10.18.1 Deviation, narrow band. The transmitter shall be operated under appropriate standard conditions. The deviation shall be measured at dial frequencies of 30.00, 52.00, 54.00, and 75.00 MHz, to determine compliance with 3.10.18.1.

4.10.18.2 Deviation, wide band. The deviation of the transmitter shall be measured at dial frequencies of 30.00, 52.00, 54.00, and 75.00 MHz to determine compliance with 3.10.18.2.

4.10.19 Deviation limiting. The transmitter shall be operated under appropriate standard test conditions. The deviation of the transmitter shall be measured when using the standard modulating test voltage. The modulating voltage shall then be increased by 14 db and the deviation measured. The ratio expressed in db of the deviation so obtained shall be recorded to determine compliance with 3.10.19. The test shall be performed at 30.00 and 54.00 MHz.

4.10.20 VFO stability. With the transmitter VFO control line shorted to ground with a 50 ohm resistor, the frequency of the transmitter oscillator shall be measured with a frequency counter at room temperature and recorded. The ambient temperature shall then be lowered to -40° C, held at this temperature for one hour, and the frequency again measured. The ambient temperature shall then be raised to +65° C, held at this temperature for one hour, and the frequency again measured. Measurements shall be made at dial frequencies of 30.50, 40.50, 51.50, 53.50, 63.50, and 74.50 MHz to determine compliance with 3.10.20.

4.11 Air-seal test. The equipment shall be opened and closed again in such manner as to break and remake the seal. Immediately thereafter, the equipment, as field transported, shall be subjected to a vacuum of 1 pound per

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square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the equipment inclosure. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. The gage used for measurement of the vacuum shall be of such accuracy that any decrease in vacuum can be readily determined. Dust caps on Antenna Control, Remote and X-Mode connector receptacles shall be removed prior to test. The receiver-transmitter shall comply with 3.6.

4.12 Visual and mechanical inspection. The equipment shall be examined for the defects listed in Standard MIL-STD-252, MIL-STD-454, MIL-P-11268 and the drawings on Drawing and Data Lists SC-DL-413500 and SC-DL-414825 (see 3.4, 3.5, and 3.17).

4.13 Inspection by gages. The mechanical dimensions and electrical performance, required by the mechanical and electrical test fixture drawings, shall be measured to determine conformance with the physical and functional requirements of 3.7. When a listed dimension or performance requirement is not within specified limits, it shall be considered a major defect. The following dimensions or performance shall be gaged or measured.

a. Mechanical (Dimensions). The gage points covered by the drawing listed on the applicable gage list specified in (1) below, and the additional gage point specified in (2) below:

(1) SC-GL-57987 for Receiver-Transmitter RT-246()/VRC
SC-GL-57988 for Receiver-Transmitter RT-524()/VRC

(2) RT-246()/VRC and RT-524()/VRC.

(a) Power supply. The location of the four holes in the power supply through with the mounting screws pass. The maximum overall dimensions and location of two electrical connectors on the chasses.

(b) Power supply cavity in case. The location of four mounting screw holes. The minimum inside overall dimensions and location of two electrical connectors in the cavity.

b. Electrical performance (modules). Prior to assembly into the receiver-transmitter, modules shall be tested to determine compliance with the performance requirements specified in Drawing and Gage List SC-GL-323124 applicable to Receiver-Transmitter RT-246()/VRC and RT-524()/VRC.

4.14 500 Hour operational test. The receiver-transmitter shall be operated continuously for 500 hours at high power with an input voltage of 30.0 vdc at a ten to one (10:1) receiver-transmit ratio at an ambient temperature of 70°F. At each 100 hour interval, the tests of 4.14e, 4.14f, 4.14g and 4.14i shall be performed to determine compliance with 3.13. After the 500 hours, without realignment or replacement of parts other than adjustment of front panel controls, the following tests shall be performed to determine compliance with 3.13.

a. Loudspeaker audio output (see 4.9.1.2).

b. Headphone audio output (see 4.9.1.3).

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c. Monitor audio output (see 4.9.1.4).

d. Distortion, narrow band (see 4.9.2.1).

e. Sensitivity (see 4.9.5).

f. Squelch sensitivity, tone (see 4.9.6.1).

g. Squelch sensitivity, noise (see 4.9.6.2).

h. Catching range (see 4.9.7).

i. Power output (see 4.10.1 and 4.10.2).

j. Distortion, narrow band (see 4.10.4 and 4.10.4.1).

k. Maximum signal plus noise to noise ratio (see 4.10.13).

l. Deviation, narrow band (see 4.10.18.1).

m. Input voltage (see 4.9.11).

4.15 Electromagnetic interference. The equipment shall comply with 3.15, using the test methods and procedures of Notice 3, MIL-STD-462.

4.16 Operational inspection. The receiver shall be operated at 30.0 and 60.0 MHz with standard test conditions except that both the old and new squelch on positions shall be used with rf inputs of 0.7 and 0.5 microvolts, respectively. At each test frequency (30 and 60 MHz), a listening test shall be performed. With the transmitter operated under standard test conditions, the power output (high and low) at 30.00, 41.50, 52.95, 53.00, 64.50 and 75.95 MHz shall be measured. Frequency selection shall be in the sequence stated in the preceding sentence. With the transmitter on low power output and tuned to 75.00, 75.05, 75.10, 75.20, 75.30, 75.40, 75.50, 75.60, 75.70, 75.80, 75.90 and 75.95 MHz, transmitter voice modulation shall be heard when an approved receiver is set at the same dial frequency as the equipment being tested. Equipment frequency selection shall first be in a clockwise direction and then in a counter-clockwise direction. In addition to the above tests, high and low power output shall be measured at each megahertz, in one megahertz steps from 30.00 MHz to 75.00 MHz to determine compliance with 3.14.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Receiver-Transmitters, Radio RT-246()/VRC and RT-524()/VRC are intended for use in forward area combat vehicles. They are used in vehicular Radio Sets AN/VRC-12() and AN/VRC-43() through AN/VRC-49().

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of specification and any amendment thereto.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Type required.

d. Marking and shipping of samples.

e. Place of final inspection.

f. Technical literature required (see 3.16).

g. Quantity of running spare parts required (see 3.16).

h. First article samples are required.

6.3 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.4.2).

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 3.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Location of operational inspection and air-seal test. It is desirable that the operational inspection (4.16) and the air-seal test (4.11) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the

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extensiveness of the changes.

6.7 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: RT-524W/VRC. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.8 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractors quality control system and the quality history of the product.

6.9 Subject term (key word) listing.

Radio telephone
Short-range
Transceiver
Frequency-modulated

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APPENDIX

Production Burn-In (Debugging) Plan for
Receiver-Transmitter, Radio RT-524()/VRC and
Receiver-Transmitter, Radio RT-246()/VRC

10.0 SCOPE

This appendix defines the requirements, methods, and procedures applicable to the burn-in (debugging) of Receiver-Transmitter Radio RT-524()/VRC and Receiver-Transmitter, Radio RT-246()/VRC. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance (see 3.12.2).

20.0 APPLICABLE DOCUMENTS

The following specification forms a part of this appendix to the extent referenced herein:

MIL-STD-781 - Reliability Tests: Exponential Distribution

30.0 REQUIREMENTS

Each Receiver-Transmitter, Radio RT-524()/VRC and Receiver-Transmitter, Radio RT-246()/VRC on contract and first article shall be subjected to the burn-in process prior to Group A electrical quality conformance inspection. When the performance measurements called for herein indicate a failure, the units will be repaired as required to eliminate the failure and returned for continuation of the burn-in processing. Units submitted for acceptance shall have completed at least 42 hours of burn-in processing. Modules used for repair shall have completed at least 28 hours of failure-free operation at the end of the 42 hour unit burn-in. Burn-in shall be conducted at the high power setting.

30.1 Test procedures.

30.1.1 Performance parameters to be measured. Audio output power and rf power output of the units in burn-in processing shall be continuously monitored to detect any cessation of performance or major degradation below specified limits. When a failure is detected by the monitoring system, the unit shall be repaired and retested to the extent deemed necessary by the Government Inspector. In addition, the following tests shall be performed on each unit prior to and following burn-in processing:

Receiver tests

The tests listed below shall be conducted at input voltages of 22.0 and 30.0 vdc.

Muting	3.9.1.1	4.9.1.1
Loudspeaker	3.9.1.2	4.9.1.2
Headphone	3.9.1.3	4.9.1.3
Monitor	3.9.1.4	4.9.1.4
Distortion, Narrow Band	3.9.2.1	4.9.2.1

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Sensitivity	3.9.5	4.9.5
Squelch		
Tone	3.9.6.1	4.9.6.1
Noise	3.9.6.2	4.9.6.2

Transmitter tests

The tests listed below shall be conducted at input voltages of 22.0 and 30.0 vdc with the exception of Automatic Frequency Selection.

	Req Para	Insp Para
Power Output		
High	3.10.1	4.10.1
Low	3.10.2	4.10.2
Automatic Frequency Selection (RT-246()/VRC only) at 25.5 vdc only	3.10.8	4.10.8
Frequency Stability	3.10.11	4.10.11
Maximum signal plus noise to noise	3.10.13	4.10.13
Deviation, Narrow Band	3.10.18.1	4.10.18.1

30.2 Environmental and operational conditions.

30.2.1 Temperature test conditions. All cycling and testing shall be done at room ambient temperature except that the room ambient temperature shall not be lower than +70° F.

30.2.2 Vibration cycling. Vibration cycling shall be in accordance with test level "B" of MIL-STD-781. (2.2G ± 10% at any non-resonant frequency between 20 and 60 Hz measured at the equipment mounting points and applied for 10 minutes out of each hour of "ON" time.)

30.2.3 Input voltage cycling. The input voltage shall be varied in accordance with the following:

Nominal voltage (25.5v)	-	50% of the time
High voltage (30v)	-	25% of the time
Low voltage (22v)	-	25% of the time

This cycling procedure shall be repeated continuously throughout the burn-in process.

30.2.4 ON-OFF cycling. All units in burn-in processing shall be subjected to ON-Off cycling at least once during every 4 hour period and shall remain in the OFF condition for at least 5 minutes but not more than 15 minutes. Time spent in the OFF condition shall not be used in calculating the total burn-in processing time specified in 30.0.

30.3 Applicable conditions.

30.3.1 Test scheduling. Testing shall be done on a 24 hour,

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7 days per week basis.

30.3.2 Duty cycle. The radio set shall be operated on a 10 minute duty cycle of 9 minutes receive and 1 minute transmit at the high setting (see 3.10.1).

30.3.3 Maintenance. No adjustment or repair shall be made on the radio set except that necessary to enable a failed radio set to perform as specified. Operator controls shall be adjusted as required.

30.4 Burn-in test plan. A detailed burn-in test plan shall be submitted for approval as specified in the contract. This test plan shall include, but not be limited to the following:

Detailed test procedures.
List of commercial test equipment used.
Schematics and complete functional descriptions of special test equipment.
Block diagrams of equipment interconnections

30.5 Documentation failure reporting and analysis. Records shall be maintained on all units and modules to demonstrate to the Government Inspector that units presented for acceptance have complied with this document. In addition, a monthly report shall be submitted to the Government activity indicated in the bid request and contract (if required), categorizing all failures by types, numbers and cycle, and furnishing an analysis as to the reasons for the failures and corrective actions taken or recommended. This report shall be submitted no later than 30 days after the month covered and shall be furnished in five copies.

40.0 PERFORMANCE

40.1 Test procedures. Each unit under test shall be subjected to the testing specified in 30.1.1. Testing shall be conducted in accordance with the applicable referenced test paragraphs and each unit shall meet the limits specified in the applicable referenced requirement paragraph. Testing shall be done under the Standard Test Conditions of 4.8 unless otherwise specified herein. Repeat all tests at 22.0 volts input voltage.

40.1.1 Special test equipment required.

Monitoring console	-	Capable of detecting failure in Audio Output Power and rf Power Output.
Vibration Tables	-	Capable of operating in accordance with MIL-STD-781.

40.2 Environmental and operational conditions. The Receiver-Transmitter (enclosed in the RT case) shall be subjected to the following conditions.

40.2.1 Vibration cycling. Vibration cycling shall be in accordance with 30.2.2 and MIL-STD-781 (Test Level B). The vibration table shall be located within the heat chamber.

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40.2.2 Input voltage cycling. A 20 ampere regulated power supply shall provide the input voltages of 22-30 volts dc as specified in 30.2.3.

40.2.3 ON-OFF cycling. ON-OFF cycling, as specified in 30.2.4, shall be accomplished by removing input power to all units.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
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3. DOCUMENT TITLE: RECEIVER-TRANSMITTER, RADIO RT-246()/VRC and RADIO RE-524()/VRC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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